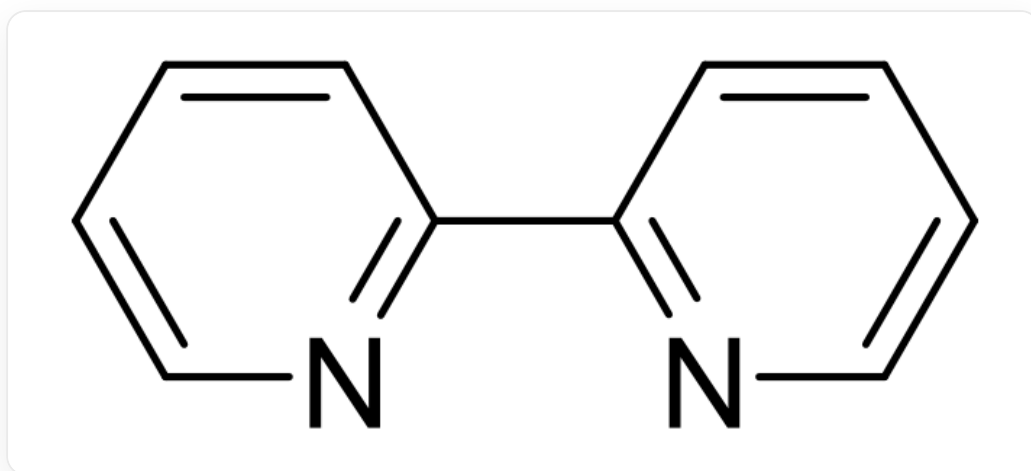


Question

X is a common **M**-containing compound, which can be obtained by treating an anhydrous ether solution of MCl_3 with hydrogen sulfide at low temperature (reaction **1**). **X** is readily soluble in potassium cyanide solution, and the products contain two different sulfur-containing anions (reaction **2**).

Y is a neutral complex containing low-valent **M**. Refluxing **Z** (containing **M** 68 %), $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, and **L** in anhydrous acetonitrile solvent for 5 min, followed by crystallization of the resulting solution under an inert atmosphere, yields the complex **Y** (reaction **3**). The structure of **L** is known as:



C1=CC(=NC=C1)C2=NC=CC=C2, i.e., the structure of [L]

In **Y**, both Zn and **M** exhibit only one coordination number, and no mirror plane is present. Elemental analysis of **Y** yields: N 12.8; C 32.9; Zn 7.46 (all in mass fraction, %).

Which of the following statements are correct?

1. Reaction **1** is a redox reaction, and the oxidation state of **M** remains unchanged in reaction **2**.
2. In the simplest balanced equation for reaction **1**, the sum of all coefficients on the reactant side and the product side is 14.
3. In reaction **2**, the mass fraction ratio of sulfur in the two anions is 1: 0.55.

4. In the simplest balanced equation for reaction **2**, the sum of all coefficients on the reactant side and the product side is 12.
5. In the simplest balanced equation for reaction **3**, the sum of all coefficients on the reactant side and the product side is 13.
6. **Y** is chiral.
7. **Y** has a total of 4 reasonable stereoisomers.

A. 0

B. 1

C. 2

D. 3

E. 4

F. 5

G. 6

H. 7

Answer

Correct Answer: **D**

Detailed Explanation

Analysis of the mass fractions of **Y**:

$$n(\text{N}) : n(\text{Zn}) = \frac{12.8}{14.01} : \frac{7.46}{65.41} = 8 : 1$$

$$n(\text{C}) : n(\text{Zn}) = \frac{32.9}{12.01} : \frac{7.46}{65.41} = 24 : 1$$

CHECKPOINT

1 PTS

In **M**, $n(\text{C}) : n(\text{N}) : n(\text{Zn}) = 24 : 8 : 1$

The chemical formula of **L** is $\text{C}_{10}\text{H}_8\text{N}_2$, containing 10 C and 2 N. Assuming only one Zn, there should be 2 **L** molecules, leaving 4 C and 4 N. Considering CN^- , the molecular weight of **Y** can be calculated:

$$Mr(\text{Y}) = \frac{65.41}{7.46\%} = 876.8$$

After subtracting one Zn and two **L** ($Mr = 156.184$), 499.022 remains. Assuming the remainder is 4 CN^- , 394.942 is left, corresponding to two Au. Examining the oxidation states: Zn is +2, so Au is +1, which aligns with chemical principles. Therefore, **M** is Au, and the chemical formula of **Y** is $\text{ZnL}_2[\text{Au}(\text{CN})_2]_2$.

CHECKPOINT

1 PTS

M is Au

CHECKPOINT

1 PTS

The chemical formula of **Y** is $\text{ZnL}_2[\text{Au}(\text{CN})_2]_2$

Knowing **M** is Au and **Y** contains CN^- , the chemical formula of **Z** (containing **M** 68 %) can be deduced as $\text{KAu}(\text{CN})_2$.

CHECKPOINT

1 PTS

The chemical formula of **Z** is $\text{KAu}(\text{CN})_2$

Thus, reaction **3** is:



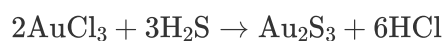
CHECKPOINT

1 PTS

Reaction **3**: $2\text{K}[\text{Au}(\text{CN})_2] + \text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O} + 2\text{L} \rightarrow \text{ZnL}_2[\text{Au}(\text{CN})_2]_2 + 2\text{KNO}_3 + 6\text{H}_2\text{O}$

The total coefficient sum is 14, so option 5 is incorrect.

For reaction **1**:



CHECKPOINT

1 PTS

Reaction **1**: $2\text{AuCl}_3 + 3\text{H}_2\text{S} \rightarrow \text{Au}_2\text{S}_3 + 6\text{HCl}$

Thus, **X** is Au_2S_3 . The sum of all coefficients on the reactant and product sides in the simplest integer equation for reaction **1** is 12, so option 2 is incorrect.

X readily dissolves in potassium cyanide solution, producing two different sulfur-containing anions (reaction **2**): S^{2-} and SCN^- . The mass fraction of sulfur in S^{2-} is 100 %, while in SCN^- it is 55 %, so option 3 is correct.

CHECKPOINT

1 PTS

The two anions produced in reaction **2** are S^{2-} and SCN^-

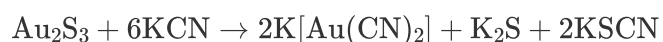
Since the oxidation state of sulfur changes, the oxidation state of **M** must also change in reaction **2**, whereas it remains unchanged in reaction **1**. Thus, option 1 is incorrect.

CHECKPOINT

1 PTS

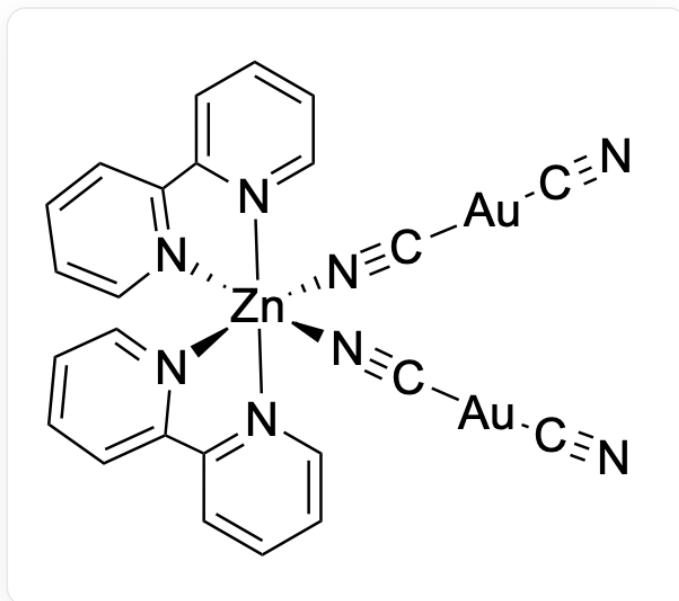
The oxidation state of **M** does not change in reaction **1** but changes in reaction **2**

The equation for reaction **2** can be written as:



The sum of all coefficients on the reactant and product sides in the simplest integer equation for reaction **2** is 12, so option 4 is correct.

From **Y**, where Zn and M each have only one coordination number and no mirror plane, the structure of **Y** is:



N#[Au]C#[N][Zn]1([N]2=C3C=CC=C2)([N]#C[Au]C#N)([N]4=CC=CC=C43)[N]5=CC=CC=C5C6=[N]1C=CC=C6

CHECKPOINT

2 PTS

The structure of **Y** is N#[Au]C#[N][Zn]1([N]2=C3C=CC=C2)([N]#C[Au]C#N)([N]4=CC=CC=C43)[N]5=CC=CC=C5C6=[N]1C=CC=C6

Clearly, **Y** is chiral and has 3 reasonable stereoisomers. Option 6 is correct, and option 7 is incorrect.

CHECKPOINT

1 PTS

Y is chiral

CHECKPOINT

1 PTS

Y has 3 reasonable stereoisomers

In summary, 3 options are correct (3, 4, 6), so the answer is D.